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Benchmarking the accuracy, robustness and reproducibility of spatial domain identification methods

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Abstract

Methods for spatial domain identification are usually compared by small differences in adjusted Rand index (ARI), mostly on one tissue and from a single run. We benchmarked the accuracy, robustness and reproducibility of 13 published methods and three training-free baselines on 20 annotated sections from three technologies (10x Visium, STARmap and MERFISH). The highest mean ARI was reached by linear diffusion on DLPFC, BANKSY $\lambda = 0.8$ on STARmap and BASS on MERFISH; a training-free diffusion baseline had a higher mean ARI than 14 of 16 methods (95% hierarchical-bootstrap intervals over donors and datasets excluding zero) and could not be separated from BASS and BANKSY $\lambda = 0.8$; no method was equivalent to it within an ARI margin of 0.03. On fixed embeddings, refitting the same mixture model from different random starts changed ARI by 0.18 on average (up to 0.60), and merely shuffling the order of the spots changed the ARI of R mclust by up to 0.37. Choosing the best of 20 starts by likelihood improved 6 of 12 methods and worsened 5. With its official code, HiSTaR reached a median ARI of 0.51 on DLPFC instead of the reported 0.65, and the released code of one method selects its reported epoch using the labels (ARI inflated by 0.147). All methods were run with default or published settings. We give recommendations for reporting spatial domain benchmarks. Code, per-run results and figures: [TODO: GitHub URL and Zenodo DOI].

Key words: spatial transcriptomics, spatial domains, clustering, reproducibility, benchmarking

Introduction

Spatial transcriptomics measures gene expression at thousands of locations in a tissue section while retaining their coordinates. Identifying *spatial domains*, regions with coherent expression such as cortical layers, is one of the first analyses of such data, and many methods have been proposed for it, from Bayesian models [Zhao et al., 2021] and neighbourhood-augmented features [Singhal et al., 2024] to graph neural networks [Hu et al., 2021, Dong and Zhang, 2022, Long et al., 2023] and hierarchical graph variational autoencoders such as HiSTaR [Yu et al., 2025].

Most of these methods are evaluated on the twelve manually annotated human dorsolateral prefrontal cortex (DLPFC) sections [Maynard et al., 2021] and report median ARI improvements of a few hundredths over previous methods. Many share the same last step: the learned embedding is clustered once with mclust's EEE Gaussian mixture [Scrucca et al., 2016]. Benchmarks have found that rankings depend strongly on the dataset [Yuan et al., 2024, VERIFY, 2025a], that deep methods vary with the random seed [VERIFY, 2025a], that replacing each method's clustering algorithm changes results [Hu et al., 2024], and that preprocessing and clustering choices often matter more than architectural novelty [VERIFY, 2026]. Simple training-free pipelines are also competitive [VERIFY, 2025b, Do et al., 2026].

We benchmark 13 published methods and three training-free baselines along three axes. *Accuracy*: how do the methods compare across three technologies, and does the ranking depend on how the embeddings are clustered? *Robustness*: on a *fixed* embedding, how much does ARI change with the initialisation of the mixture model and with perturbations that leave the data essentially unchanged (spot order, negligible noise), and does selecting fits by likelihood help? *Reproducibility*: do published results reproduce with the official code (here HiSTaR's DLPFC results), and does released code select models using the labels? The study design is summarised in Fig. 1.

Materials and methods

Data

DLPFC (10x Visium). Twelve sections from three donors [Maynard et al., 2021]; seven annotated layers (five for donor 2), `layer_guess_reordered`. **STARmap mouse medial prefrontal cortex.** Three sections (about 1050 cells, 166 genes) with

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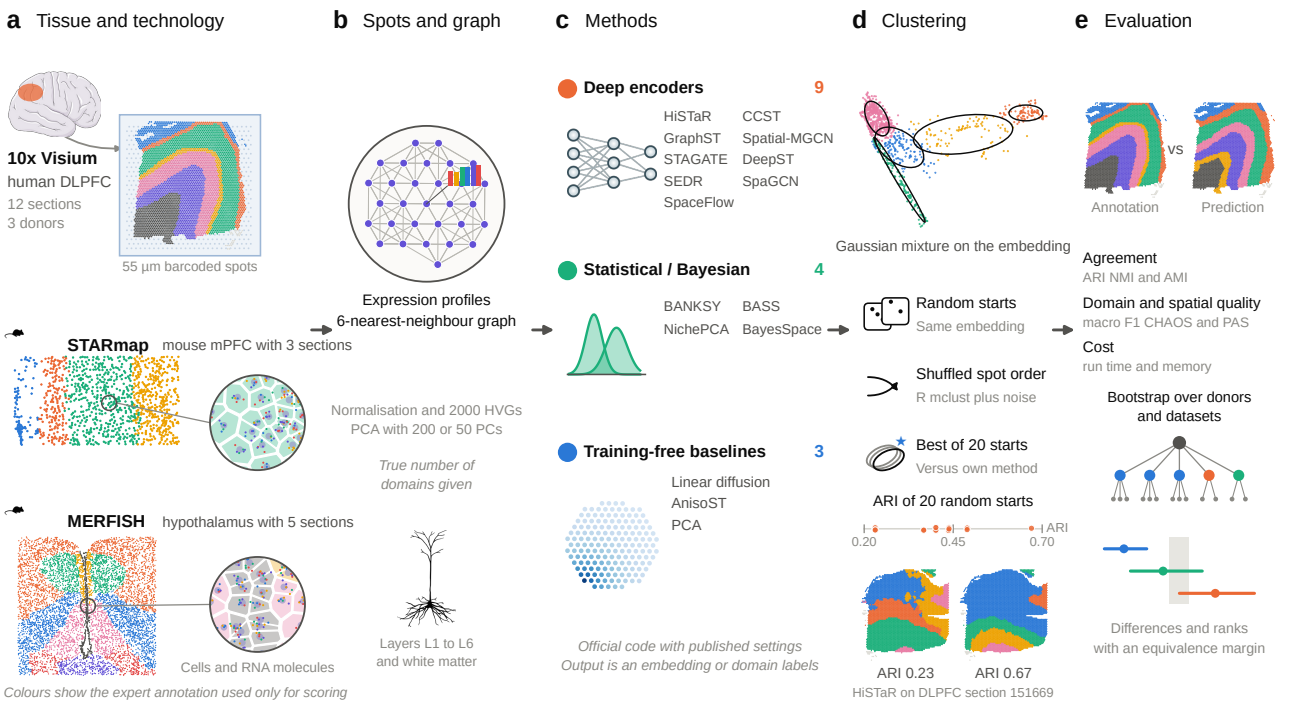


Fig. 1. Study design. (1) Human DLPFC on a 10x Visium capture area (section 151673) and mouse sections profiled by STARmap (mPFC, section BZ5) and MERFISH (hypothalamus, Bregma −0.14); colours are the expert annotation, used only for scoring. Circles zoom into individual cells (Voronoi cells from the measured positions, with RNA molecules drawn schematically). (2) Each spot or cell carries a gene-expression profile and is linked to its six nearest neighbours; the true number of domains is given to every method. (3) 13 published methods and three training-free baselines in three families (colours used throughout). (4) Each embedding is clustered from 20 random starts, with R mclust after shuffling the spot order or adding noise, and with a 20-start best-likelihood fit compared with the method's own clustering. Below: one fixed HiSTaR embedding of DLPFC section 151669 clustered from 20 random starts (dots: ARI of each start) and the domains of the worst and best start. (5) Predicted domains are compared with the annotation (here linear diffusion on 151673) using agreement, per-domain, spatial and cost metrics, summarised with a bootstrap over donors and datasets. Icons: Bioicons (CC0) and Servier Medical Art (CC BY 3.0).

four annotated layers (L1, L2/3, L5, L6) [Wang et al., 2018]. **MERFISH mouse hypothalamus.** Animal 1, five sections from Bregma −0.04 to −0.24 (about 5500 cells, 155 genes) with eight annotated regions [Moffitt et al., 2018]. The STARmap and MERFISH annotations are those distributed with the BASS analyses [Li and Zhou, 2022]. In total, 20 sections from three technologies (Supplementary Table S1; Supplementary Note 1).

Preprocessing

DLPFC: genes in fewer than 50 spots or with fewer than 10 counts were removed, 2000 highly variable genes were selected (Seurat v3 flavour), counts were library-size normalised, log-transformed and scaled, and reduced to 200 principal components (PCs). STARmap and MERFISH (targeted panels): all genes, same normalisation, 50 PCs. All methods received the same input where they accept one; BANKSY used the log-normalised genes, and SpaGCN, GraphST and the methods added later (below) their own documented preprocessing of the same raw counts. The true number of domains was given to every method, and unannotated spots were excluded from evaluation.

Methods

- **HiSTaR** [Yu et al., 2025]: PyTorch port of the official code with the published defaults (cross-level weight 0.3, 200+200 epochs).
- **GraphST** [Long et al., 2023]: official package and tutorial pipeline (600 epochs, 20-PC embedding, R mclust EEE; with and without its label refinement). Because of its CPU cost, GraphST was run with one seed on DLPFC and three seeds on STARmap and MERFISH.
- **BANKSY** [Singhal et al., 2024]: official `pybanksy`, 18 neighbours, $m_{\max} = 1$, $\lambda \in \{0.2, 0.8\}$, 20 PCs.
- **SpaGCN** [Hu et al., 2021]: official tutorial pipeline without histology ($p = 0.5$, Louvain resolution search to the true number of domains, 200 epochs).
- **Methods added in the extended study**, each run with its official code and tutorial settings on one CPU thread: STAGATE [Dong and Zhang, 2022] (R mclust), SEDR [VERIFY, 2024] (R mclust), SpaceFlow [VERIFY, 2022b] (Leiden with a search for the true number of domains), NichePCA [VERIFY, 2025b] (Leiden, same search), CCST [VERIFY, 2022a] and Spatial-MGCN [VERIFY, 2023] (both k-means), DeepST [VERIFY, 2022c] (Leiden with refinement), BASS [Li and Zhou, 2022] and BayesSpace [Zhao et al., 2021] (Bayesian models that cluster internally; BayesSpace requires a spot lattice and was run on the Visium sections only). NichePCA, SEDR, SpaceFlow and DeepST were run with three seeds per section; STAGATE,

CCST, Spatial-MGCN and BASS with one seed (seed 0) and BayesSpace once per section. Deviations from the tutorials (for example the neighbourhood radius, expressed in spot spacings so that it transfers across technologies) are listed in Supplementary Table S2. Spatial-MGCN's released test script reports the epoch with the highest ARI against the annotation; in the benchmark we report its final epoch (label-free) and give the label-selected value separately.

- **Training-free baselines.** *Linear diffusion*: the top 30 PCs are smoothed on the spatial 6-nearest-neighbour graph, $H^{(t+1)} = (1-\alpha)H^{(t)} + \alpha D^{-1}AH^{(t)}$, $\alpha = 0.5$, $T = 10$. *AnisoST*: the same with edge weights recomputed at every step, $w_{ij}^{(t)} = \exp(-\|h_i^{(t)} - h_j^{(t)}\|^2/\kappa^{(t)})$ (Perona–Malik-type edge-preserving diffusion [Perona and Malik, 1990]; $\kappa^{(t)}$ the median squared edge difference); a similar idea has been used inside a deep model [Wang et al., 2024]. *PCA* without smoothing is the non-spatial reference. All baseline settings were fixed on DLPFC donor 1 before any other data were analysed, and none were tuned on STARmap or MERFISH; the sensitivity of linear diffusion to each setting is shown in Supplementary Fig. S2. All methods were run with default or published settings; none were tuned per dataset (equal tuning budgets are a topic for future work).

Clustering protocols

Standard: each method's usual clustering. This is R mclust for GraphST, STAGATE and SEDR, a 3-start scikit-learn GMM [Pedregosa et al., 2011] with a fixed seed for HiSTaR, BANKSY and the baselines, Leiden or k-means for the other methods, and the internal model for SpaGCN, BASS and BayesSpace. *Robust*: the method's embedding clustered with a tied-covariance Gaussian mixture (the mclust EEE model) with 20 random starts, keeping the fit with the highest log-likelihood (label free); available for every method that outputs an embedding. *Lottery*: one seed-0 embedding per section and method, clustered 20 times with a single random start each (14 methods with embeddings). *mclust perturbations*: R mclust (version 6.0.1, default hierarchical initialisation) on the same embedding (i) unchanged, (ii) after random permutation of the rows, 10 times, and (iii) after adding Gaussian noise with 1% of each feature's standard deviation, 10 times (14 methods in the extended set, plus HiSTaR, GraphST and AnisoST).

Evaluation and statistics

ARI [Hubert and Arabie, 1985] against the manual annotation; NMI gave the same conclusions (Supplementary Table S4); further agreement, spatial and label-free metrics are in Supplementary Table S12 (definitions in Supplementary Note 3). For every (method, section) we average over seeds. Sections are not independent (12 DLPFC sections come from three donors; the STARmap and MERFISH sections come from one experiment each), so the unit of analysis is the cluster: 5 clusters (three DLPFC donors, STARmap, MERFISH). For each method we estimate the difference in mean ARI from linear diffusion, the reference, which was chosen after the means were seen, so this comparison is exploratory. Intervals come from a hierarchical bootstrap (5000 resamples of clusters, then of sections within a cluster; the estimate is the equal-weight mean of cluster means) with 95% percentile intervals (Supplementary Note 4). A method is called equivalent to the reference if the 90% interval lies within ± 0.03 ARI, a margin fixed before the analysis. Ranks and the probability of ranking first come from the same bootstrap. The analysis is repeated without the development sections (DLPFC donor 1 and the first section of STARmap and MERFISH). Run time and peak memory were measured for a subset of methods in a standardised runner (process-tree sampling, one CPU thread per job; Supplementary Note 6). Supplementary Table S18 lists which method was run in which experiment, with the number of sections and seeds.

Results

Accuracy across methods and technologies

Fig. 2 and Fig. 3 compare all methods under both protocols (per-dataset values in Supplementary Table S3); Fig. 4 shows example partitions. Across all methods and both protocols, the highest mean ARI was reached by linear diffusion on DLPFC (0.558), BANKSY $\lambda = 0.8$ on STARmap (0.740) and BASS on MERFISH (0.582). Linear graph diffusion had a mean ARI of 0.558, 0.705 and 0.497 on the three datasets (average over datasets 0.587, against 0.594 for BASS and 0.554 for BANKSY $\lambda = 0.8$). Because sections are nested in 5 donors and datasets, we compared methods with a hierarchical bootstrap (Fig. 2b; Supplementary Table S5). With each method's standard clustering, linear diffusion had a higher mean ARI than 14 of 16 methods (95% interval above zero), and this held for 12 of them with the development sections excluded. Two comparisons were unresolved: BASS (Δ +0.025, 95% interval −0.046 to +0.101) and BANKSY $\lambda = 0.8$ (+0.045, −0.008 to +0.091). Neither was equivalent to the baseline within an ARI margin of 0.03, and the per-dataset differences changed sign: BASS was ahead on MERFISH and BANKSY $\lambda = 0.8$ on STARmap. With 20-start clustering, 8 of 13 comparisons favoured the baseline and 5 were unresolved (Supplementary Table S6). In the bootstrap, linear diffusion ranked first in 0.70 of the resamples (95% rank interval 1–3 among 16 methods) and BASS in 0.25 (Fig. 2c; Supplementary Table S7). These intervals rest on 5 clusters and are wide, and the reference was chosen after seeing the means, so the comparison is exploratory. The ranking of the deep methods relative to the baselines changed with the clustering protocol (Fig. 3). Seed-to-seed variability also differed: HiSTaR's standard deviation of ARI was 0.047 on DLPFC and 0.18 on STARmap, against 0.003–0.04 for the training-free baselines (Supplementary Table S8). Run time and memory differed by orders of magnitude between methods (Supplementary Table S13); in a separate feasibility test, one DLPFC section took tens of minutes on one CPU thread for most of the neural methods and about 50 minutes for BayesSpace.

Edge preservation helps at boundaries on Visium but not on single-cell data

On DLPFC, AnisoST matched linear diffusion in ARI (0.551 vs 0.558) and improved accuracy on spots at layer boundaries by 0.025 (9/12 sections, $p = 0.034$, uncorrected) without changing interior accuracy (Supplementary Fig. S3). A pilot check that both baselines are invariant to transformations that should not change the answer is in Supplementary Table S17. On MERFISH it was clearly worse (0.363 vs 0.497): at single-cell resolution neighbouring cells often differ by cell type, so an edge-preserving rule blocks the smoothing that is needed to reveal regions. We report this as a negative result for the edge-preserving variant; the simpler linear diffusion was the more robust baseline.

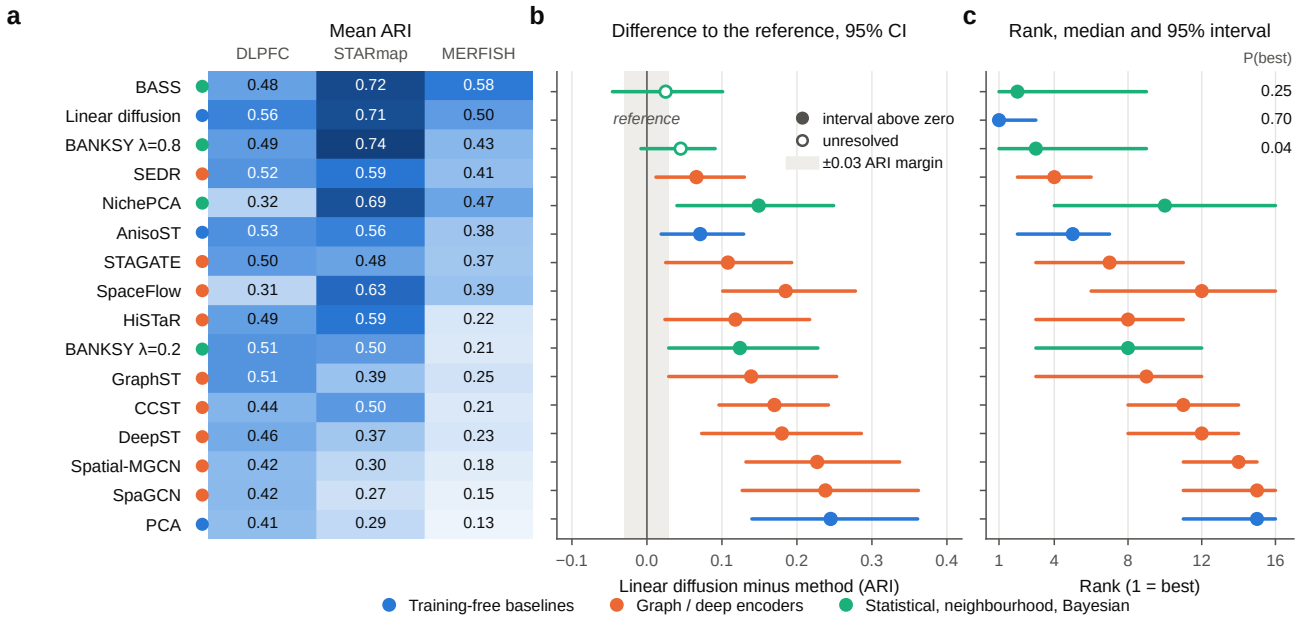


Fig. 2. Accuracy of 13 published methods and three training-free baselines (BayesSpace, run on DLPFC only, is omitted). **a**, Mean ARI per dataset with each method's standard clustering (training-free baselines: their 20-start clustering). Rows are ordered by the mean over the three datasets. Dots mark the method family. **b**, Difference in mean ARI between linear diffusion and each method, with 95% intervals from a hierarchical bootstrap over 5 clusters (three DLPFC donors, STARmap, MERFISH). Filled symbols: interval above zero; open symbols: unresolved. The grey band is the equivalence margin of ±0.03 ARI. **c**, Rank of each method in the bootstrap (dot: median; line: 95% interval); numbers give the probability of ranking first. Linear diffusion was chosen as the reference after seeing the means, so this comparison is exploratory.

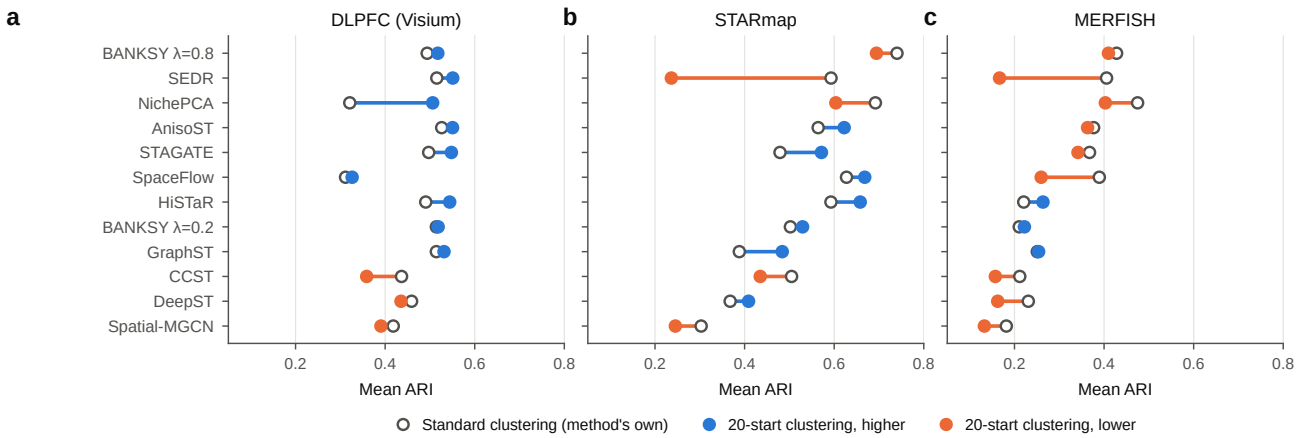


Fig. 3. Effect of the clustering protocol on mean ARI. Open circles: each method's standard clustering. Filled circles: the 20-start, best-likelihood Gaussian mixture applied to the same embedding (blue: higher, orange: lower). Methods that cluster internally (SpaGCN, BASS, BayesSpace) have no embedding and are not shown.

Robustness to the clustering initialisation

Figure 5a shows the ARI of 20 single-start fits of the same mixture model on each fixed embedding. Across the 280 embeddings of 14 methods, the ARI range within an embedding was 0.18 on average and reached 0.60 (SEDR, dlpfc 151672). The effect depended on the method: the mean range was 0.09 for SpaceFlow and 0.24 for BANKSY $\lambda = 0.8$ (Fig. 5a); for HiSTaR and AnisoST it was 0.23 and 0.20, whereas GraphST's 20-dimensional PCA embeddings were more stable (0.09). The best of the 20 fits by ARI exceeded the average single fit by 0.08: choosing a clustering run by its agreement with the labels, explicitly or by trying a few seeds, would inflate a reported result by about as much as the gap between most published methods. Per-method ranges are in Supplementary Table S9 and every fit is shown in Supplementary Fig. S1.

Robustness to spot order and noise in R mclust

mclust initialises EM by model-based hierarchical agglomeration, so one call on a fixed embedding always returns the same partition. That partition, however, depended on details that carry no information (Fig. 5b). Over 280 embeddings, randomly permuting the order of the spots changed ARI by 0.10 on average within an embedding (up to 0.37); on DLPFC the mean was 0.12 and on MERFISH 0.11. In 26% of the permutations

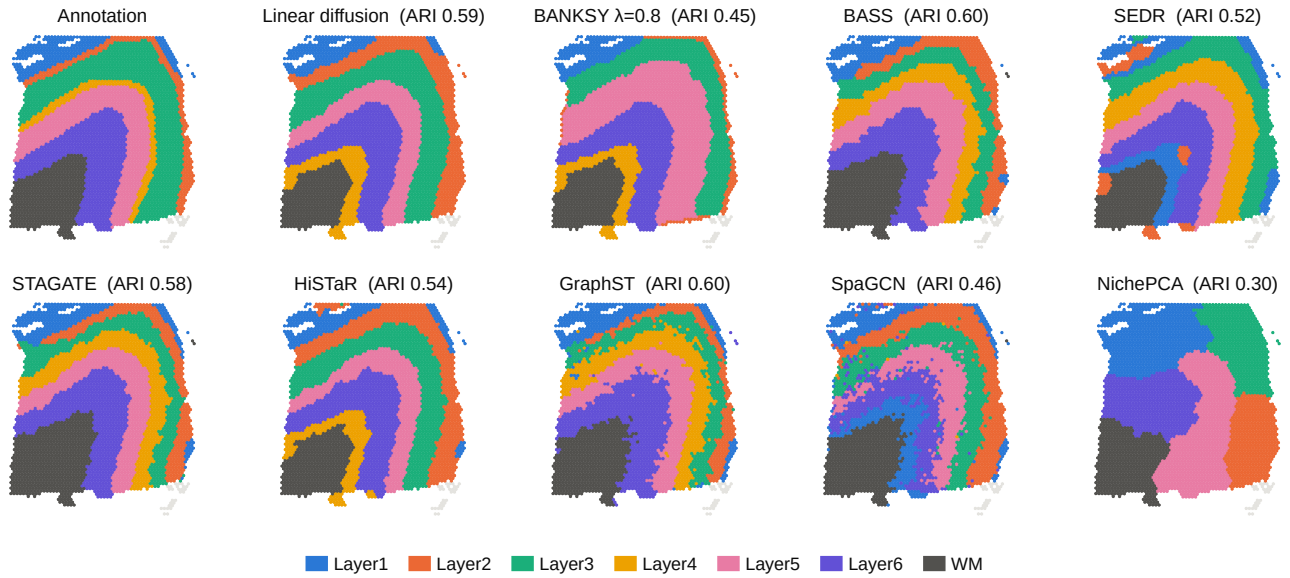


Fig. 4. Domains on DLPFC section 151673 (seed 0). The panels show the manual annotation and the partition of nine methods; predicted clusters were matched to the annotated layers with the Hungarian algorithm for display only, and the ARI against the annotation is given in each title. Spots without annotation are light grey.

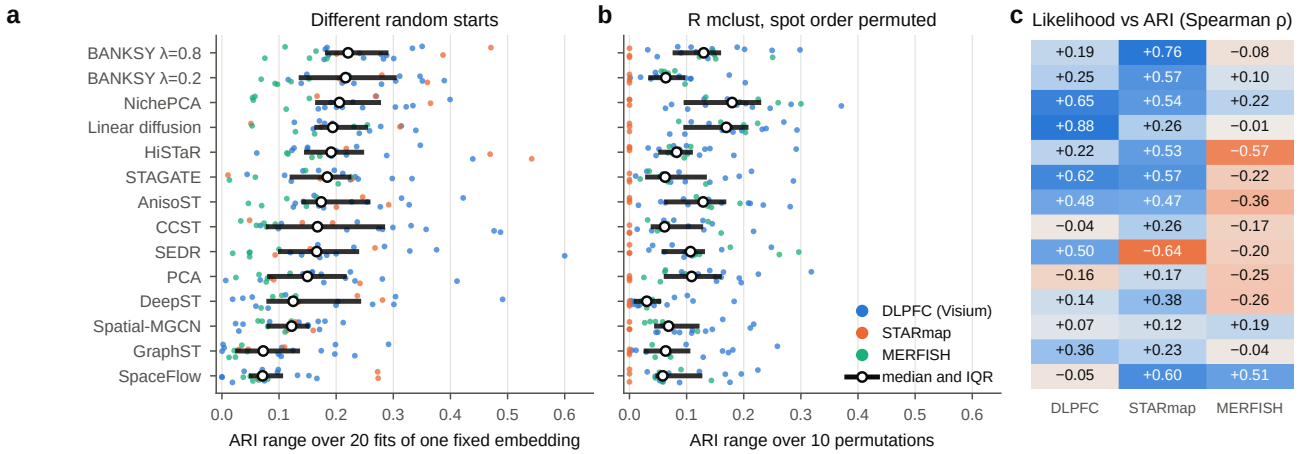


Fig. 5. Robustness of the clustering step on fixed embeddings (14 methods with embeddings; one dot per section; colour: dataset). **a**, ARI range over 20 single-start fits of the same Gaussian mixture model on one embedding. **b**, ARI range of R mclust (default settings) over 10 random permutations of the row order of the same embedding. Black bars: interquartile range; open circle: median. Methods are ordered by the median range in **a**. **c**, Median Spearman correlation between the log-likelihood and the ARI of the 20 fits (positive: better-fitting mixtures agree better with the annotation).

ARI moved by more than 0.05, and for 44% of the embeddings the range exceeded 0.1. Adding noise at 1% of each feature's standard deviation had similar effects (mean range 0.13, up to 0.56; 0.22 on STARmap). Over these embeddings, mclust's default fit had a mean ARI of 0.40 against 0.43 for the best-likelihood fit of 20 random starts, and its log-likelihood was at least as high in only 34% of cases (Supplementary Table S10).

Choosing fits by likelihood is not a remedy

A natural fix is to run many starts and keep the most likely fit. It helped some methods and hurt others. Replacing each method's standard clustering by this matched protocol raised mean ARI for 6 of 12 methods and lowered it for 5. For example, it raised HiSTaR's mean ARI by +0.053 (14/20 sections, $p = 0.004$) and GraphST's by +0.025 (13/20, $p = 0.114$), but changed SEDR's by −0.092 and Spatial-MGCN's by −0.037 (Supplementary Table S11). The reason is visible in Fig. 5: likelihood and ARI were positively related on DLPFC (median Spearman ρ 0.38) but not on MERFISH (median ρ −0.11; negative for 10 of 14 methods), where the best-fitting mixture can separate cell types rather than anatomical regions. Better optimisation of the mixture model is therefore not the same as better domains.

Reproducibility of HiSTaR's published DLPFC results

With the official code and defaults, HiSTaR reached a median ARI over the twelve DLPFC sections of 0.509 (mean 0.491, 95% bootstrap CI 0.454–0.525) with the 3-start GMM, and 0.449 (mean, seed 0) when its embedding was clustered with

R mclust as in the official pipeline, compared with a reported median of 0.65 [TODO: verify the reported values (0.65 median; 0.718 on 151673) against the HiSTaR article]. Fixing a bug that applies random masking at inference did not change accuracy (0.487; Supplementary Table S14). Possible causes include per-section tuning of the loss weights (described in the HiSTaR supplement), unpublished preprocessing and the choice of seed; the robustness results above show that the clustering step alone produces differences of this size. [TODO: Report the outcome of contacting the authors.]

Reproducibility and label use in released code

[TODO: Contact the Spatial-MGCN authors and report their response before including this subsection.] Spatial-MGCN's released test scripts (`DLPFC_test.py`, `HBC_test.py`) compute ARI against the annotation at every training epoch and keep the best epoch for reporting. Applying the same selection rule in our runs (seed 0, 20 sections), this label-selected epoch had a mean ARI of 0.551 on DLPFC, 0.687 on STARmap and 0.221 on MERFISH, against 0.418, 0.303 and 0.182 for the final epoch used in our benchmark, i.e. an inflation of 0.147 on average (0.38 on STARmap; Supplementary Table S15). Selecting a run, epoch or seed by agreement with the labels is a form of label leakage that makes a method look competitive with the training-free baselines (0.551 against 0.558 on DLPFC) when its label-free result is not. An automated screen of all released repositories for other patterns of label use, with every candidate line read by hand, is reported in Supplementary Table S16 and Supplementary Note 5.

Discussion

We benchmarked 13 published methods and three training-free baselines on 20 sections from three technologies. Accuracy differences between most methods were small, and a training-free diffusion baseline matched or outperformed every deep encoder we tested. Robustness was a larger concern: on fixed embeddings from 14 methods, random initialisation of the clustering step alone spanned 0.18 ARI on average, the R implementation used by most published methods changed its answer when the spots were merely reordered, and better likelihood did not reliably mean better domains. Reproducibility was limited by both the clustering step and by choices in released code, including selecting a reported epoch with the labels. These results agree with recent large benchmarks [VERIFY, 2026, Hu et al., 2024, VERIFY, 2025a] and add a direct, quantitative measurement of clustering variance on fixed embeddings across a broader set of methods.

Recommendations.

(1) Report the clustering protocol: algorithm, implementation and version, initialisation, number of starts, seeds, and how they were chosen. (2) Never select seeds, epochs or runs by agreement with the labels. (3) Report distributions over training seeds *and* clustering runs (or input permutations), with per-section values. (4) Compare encoders under the same clustering, and include training-free baselines such as neighbourhood smoothing. (5) Report runtime, memory and seed variance alongside accuracy.

Limitations.

All methods were run with default or published settings (we did not tune any method and we could not reproduce HiSTaR's per-section tuning), so our results do not show what each method can reach with its authors' own tuning. Several methods were run with one seed (Supplementary Table S18), and BayesSpace on Visium only. The lottery and mclust experiments use one embedding per section and method. The STARmap and MERFISH annotations are coarse and the STARmap data contain only three small sections. Three technologies and 20 sections do not cover the breadth of spatial data (for example Slide-seq, Stereo-seq or Xenium). The number of domains was given to every method. Annotations may themselves have been derived with methods similar to the ones evaluated.

Competing interests

No competing interest is declared.

Author contributions statement

[TODO: CRediT statement.]

Use of AI tools

[TODO: Describe per journal policy: code, experiments and a first draft of this manuscript were produced with the assistance of an AI coding assistant (Claude, Anthropic); all results were checked by the authors, who take full responsibility for the content.]

Supplementary data

Supplementary Data are available at NAR Online.

Data availability

DLPFC: spatialLIBD (count matrices) and SEDR analyses repository (layer annotations). STARmap and MERFISH with domain annotations: BASS-Analysis repository [Li and Zhou, 2022], from the original studies [Wang et al., 2018, Moffitt et al., 2018]. Scripts that download and convert all data are provided with the code.

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[TODO: Funding.]

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